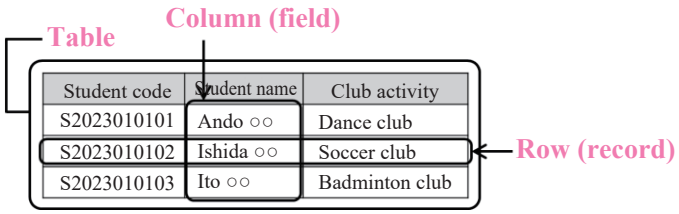
9-2 Database [2]

Point!

Relational Database

(1) ([1]**Relational database** (**RDB**)) : Database where collected data is organized and managed across multiple tables.

- A spreadsheet ([2]**table**) consists of ([3]**rows** (**records**)) and ([4]**columns** (**fields**)).



- Establishing ([5]**relationships**) between multiple tables makes it possible to eliminate duplicate data, which allows data to be handled with integrity.

(2) ([6]**SQL**) : A language used in relational databases to manipulate data.
Performs data registration, insertion, retrieval, and deletion.

(3) Relational Database Operations (Relational Algebra Operations)

[1] ([7]**Selection**) : Only rows that meet given conditions are extracted and displayed.

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Record for club code C2 → **Selection**

Student code	Student name	Club code
S2023010102	Ishida ○○	C2
S2023010105	Ezawa ○○	C2

[2] ([8]**Projection**) : Displaying only certain columns from a table.

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Student name → **Projection**

Student name
Ando ○○
Ishida ○○
Ito ○○
Udagawa ○○
Ezawa ○○

[3] ([9]**Join**) : Linking data from multiple tables according to specific conditions and displaying it as a single table.

Club table

Club code	Club activity
C1	Dance club
C2	Tennis club

Student table

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Join

Student code	Student name	Club code	Club activity
S2023010101	Ando ○○	C1	Dance club
S2023010102	Ishida ○○	C2	Tennis club
S2023010103	Ito ○○	C1	Dance club
S2023010104	Udagawa ○○	C1	Dance club
S2023010105	Ezawa ○○	C2	Tennis club

Warm Up

Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms.

> Summarizing data into a table often makes it easier to understand. A database where data is managed in a table is referred to as a ([1]). In a ([1]), rows are referred to as ([2]), and columns are referred to as ([3]). Also, establishing ([4]) between multiple tables makes it possible to eliminate duplicate data, which allows data to be handled with integrity.

(2) Consider the operations of relational databases. Table 3 below was created by performing certain operations with Table 1 and Table 2. List all the operations performed among selection, projection, and join.

Table 1

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Table 2

Club code	Club activity
C1	Dance club
C2	Tennis club

Table 3

Student code	Student name	Club activity
S2023010101	Ando ○○	Dance club
S2023010102	Ishida ○○	Tennis club
S2023010103	Ito ○○	Dance club
S2023010104	Udagawa ○○	Dance club
S2023010105	Ezawa ○○	Tennis club

Explanation

(1) [1] Relational database (RDB) [2] Record [3] Field [4] Relationships

(2) Table 3 combines “Student code” and “Student name” from Table 1, and “Club activity” from Table 2, indicating that a join operation has been performed. Also, since the “Club code” in Table 1 is not included in Table 3, it can be concluded that it also involves a projection.
Therefore, join and projection

Try

Answer the following questions.

(1) Choose the appropriate name of the database where data is managed as a table from the options **A** to **C** below, and answer using the letter.

A Hierarchical database **B** Network database

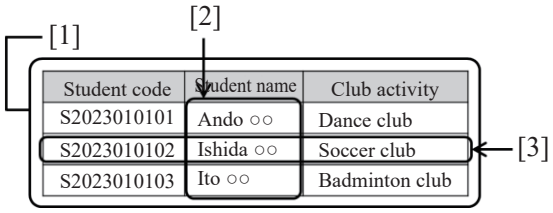
C Relational database

(2) Choose two appropriate names for each of the sections [1] to [3] in the table on the right from the options **A** to **F** below, and answer using the letters.

A Spreadsheet **B** Row

C Column **D** Field

E Table **F** Record



(3) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

Searches in a relational database are conducted using relational algebra operations.
- The process of extracting multiple columns from a table to create a new table is referred to as ([1]).
- The process of extracting only the rows that meet given conditions to create a new table is referred to as ([2]).
- The process of linking multiple tables based on the relationship between certain items to create a new table is referred to as ([3]).

A Projection **B** Join **C** Selection **D** Normalization

(4) For tables [1] to [3] which were created using the following two tables (Student table and Club table), choose the name of the operation that was performed from the options **A** to **D** below, and answer using the letters.

Student table

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Club table

Club code	Club activity
C1	Dance club
C2	Tennis club

[1]

Student name
Ando ○○
Ishida ○○
Ito ○○
Udagawa ○○
Ezawa ○○

[2]

Student code	Student name	Club code
S2023010102	Ishida ○○	C2
S2023010105	Ezawa ○○	C2

[3]

Student code	Student name	Club code	Club activity
S2023010101	Ando ○○	C1	Dance club
S2023010102	Ishida ○○	C2	Tennis club
S2023010103	Ito ○○	C1	Dance club
S2023010104	Udagawa ○○	C1	Dance club
S2023010105	Ezawa ○○	C2	Tennis club

A Projection **B** Join **C** Selection **D** Normalization

Exercise

1 Answer the following questions.

(1) Choose one correct statement regarding relational databases from the options **A** to **D** below, and answer using the letter.

A The operation of searching for and extracting records that meet specific conditions from a table is called "projection."

B A table consists of records and fields, and allows data to be managed in a table format.

C To maintain data integrity, it is not possible to link data from multiple tables.

D By displaying only specific columns from a table, it is possible to eliminate data redundancy.

(2) Choose one correct statement regarding SQL from the options **A** to **D** below, and answer using the letter.

A Mechanism that ensures the integrity of data stored in a relational database.

B Eliminating duplicate data from a relational database.

C Programming language for manipulating relational databases.

D Operations performed on a relational database such as selection, projection, and join.

(3) Choose one correct statement regarding the "join" operation of a relational database from the options **A** to **D** below, and answer using the letter.

A Creating a new table by linking information from multiple tables based on certain conditions.

B Using a language called SQL to register or delete data in a table.

C Extracting only the records from a table that meet specific conditions.

D Extracting only the fields from information contained in a table that meet specific conditions.

(4) For tables [1] to [3] which were created using the following two tables (Employee table and Branch table), choose the name of the operation that was performed from the options **A** to **D** below, and answer using the letters.

Employee table

Employee number	Name	Department	Branch number
10001	Sato ○○	Sales department	101
10002	Suzuki ○○	General affairs department	101
10003	Sakai ○○	Accounting department	102
10004	Kato ○○	Human resources department	102

Branch table

Branch number	Branch name
101	Tokyo
102	Osaka
103	Kyoto
104	Sapporo

[1]

Employee number	Name	Department	Branch number	Branch name
10001	Sato ○○	Sales department	101	Tokyo
10002	Suzuki ○○	General affairs department	101	Tokyo
10003	Sakai ○○	Accounting department	102	Osaka
10004	Kato ○○	Human resources department	102	Osaka

[2]

Employee number	Name	Department	Branch number
10003	Sakai ○○	Accounting department	102
10004	Kato ○○	Human resources department	102

[3]

Name	Department
Sato ○○	Sales department
Suzuki ○○	General affairs department
Sakai ○○	Accounting department
Kato ○○	Human resources department

A Projection **B** Join **C** Selection **D** Normalization